

“Autonomous Robotic Ultrasound for Fetal Monitoring”

A PROJECT REPORT

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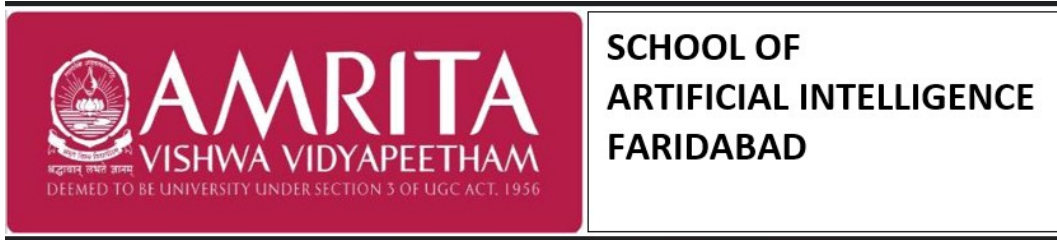


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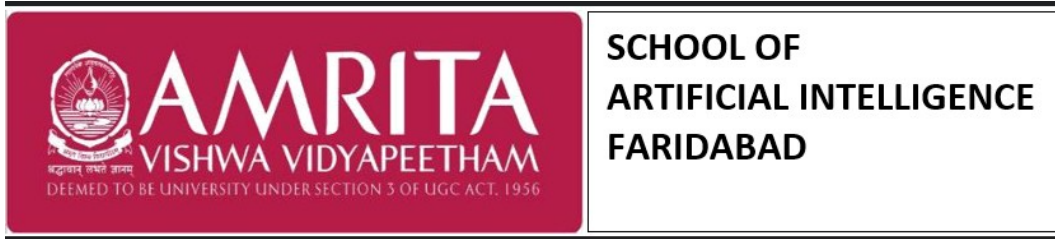
BONAFIDE CERTIFICATE

This is to certify that this project report entitled “**Autonomous Robotic Ultrasound for Fetal Monitoring**” is the **bonafide work** of the following students, who carried out the project work under my supervision:

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DECLARATION BY THE CANDIDATE

We hereby declare that the project report entitled “**Autonomous Robotic Ultrasound for Fetal Monitoring**” submitted by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology is the record of the project work carried out by us under the guidance of **Dr. Anirban Tarafdar (Mathematics for Intelligent Systems–3)** and **Mr. Jayaprakash (Introduction to Robotics)**. This work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or title in this or any other University or institution of higher learning.

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ABSTRACT

This project presents an autonomous robotic ultrasound system designed to perform fetal heartbeat detection without the need for a human sonographer. The system uses a 6-degree-of-freedom robotic arm equipped with an ultrasound probe to safely and accurately scan a pregnant woman’s abdomen, making it particularly suitable for rural and telemedicine applications where trained ultrasound operators are unavailable.

Unlike conventional position-controlled robotic systems, the proposed approach integrates force control, surface-aware kinematics, and ultrasound signal analysis. The robot maintains optimal probe pressure and orientation while moving across the curved abdominal surface, ensuring reliable acoustic coupling and patient safety. Abdominal geometry is obtained through synthetic models, 2D image-based contour extraction, or 3D point cloud reconstruction, allowing both simplified testing and patient-specific simulation.

An optimized scanning trajectory is generated over the reconstructed surface, and inverse kinematics is used to align the probe normal to the anatomy at each scan point. A force-regulated interaction model simulates soft tissue behavior, while ultrasound signal quality and fetal heart rate are estimated in real time. Scan confidence is evaluated based on probe force, orientation, and signal attenuation, enabling comparison between optimized and non-optimized control strategies.

The results demonstrate that combining force-aware control with geometry-informed trajectory planning significantly improves scan consistency and fetal heartbeat detection. This work highlights the feasibility of autonomous robotic ultrasound as a foundation for future remote prenatal diagnostic systems.

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LIST OF SYMBOLS

Symbol	Description
$\mathbf{q}$	Joint angle vector of the 6-DOF robotic manipulator
$\mathbf{T}$	Homogeneous transformation matrix
$\mathbf{R}$	Rotation matrix of the end-effector
$\mathbf{p}$	Position vector of the ultrasound probe
$\mathbf{n}$	Local surface normal vector of the abdominal surface
$\mathbf{J}$	Jacobian matrix of the robot
$\mathbf{J}^\dagger$	Moore–Penrose pseudoinverse of the Jacobian
$\mathbf{x}$	End-effector pose (position + orientation)
$\mathbf{x}_d$	Desired end-effector pose
$F(t)$	Contact force applied by the ultrasound probe at time t
$\delta(t)$	Probe indentation depth into tissue
$k(t)$	Time-varying tissue stiffness
F_{opt}	Optimized probe contact force
F_{max}	Maximum allowable safe contact force
θ	Probe misalignment angle relative to surface normal
d	Penetration depth of ultrasound into tissue
C	Ultrasound confidence metric
α	Attenuation coefficient of ultrasound waves
$p(\mathbf{x}, t)$	Acoustic pressure field
c	Speed of sound in biological tissue
∇^2	Laplacian operator
$S(\mathbf{x}, t)$	Ultrasound source term
f_{HR}	Fetal heart rate frequency
BPM	Beats per minute (fetal heart rate)
$r(t)$	Time-varying spiral trajectory radius
ω	Angular velocity of scan trajectory
(x, y, z)	Cartesian coordinates of scan points
$\mathcal{S}$	Reconstructed abdominal surface
α -shape	Surface reconstruction method for point clouds
$\mathbf{V}$	Set of surface vertices
$\mathbf{F}$	Set of triangular surface faces
N	Total number of scan waypoints
t	Discrete scan index or time step

1. Introduction

Ultrasound imaging plays a critical role in prenatal healthcare, providing a safe, non-invasive, and cost-effective means of monitoring fetal development and assessing vital indicators such as fetal heart rate. Despite its widespread use, the quality and reliability of ultrasound examinations are highly dependent on the expertise of trained sonographers, who must continuously adjust probe position, orientation, and contact pressure to obtain diagnostically useful images. In many rural and underserved regions, the shortage of skilled personnel significantly limits access to routine prenatal screening, increasing the risk of delayed diagnosis and adverse maternal-fetal outcomes. These challenges motivate the development of autonomous robotic ultrasound systems capable of performing standardized scans without requiring constant human operation.

- Ultrasound imaging serves as a critical tool in prenatal healthcare
- Provides safe, non-invasive, and cost-effective monitoring of fetal development
- Assesses vital indicators, including fetal heart rate
- Quality and reliability depend on trained sonographer expertise
- Requires continuous adjustment of probe position, orientation, and contact pressure for diagnostic imaging
- Rural and underserved regions face significant shortage of skilled personnel
- Limited access to routine prenatal screening increases risk of delayed diagnosis
- Adverse maternal-fetal outcomes are more prevalent in resource-limited settings
- Autonomous robotic ultrasound systems are needed for standardized scans without constant human operation



Figure 1: Doctor taking Ultrasound of pregnant woman.

Robotic manipulation offers inherent advantages for ultrasound acquisition, including precise motion control, repeatability, and the ability to integrate advanced sensing and feedback mechanisms. However, autonomous ultrasound scanning presents unique technical challenges due to the soft, curved, and patient-specific nature of human anatomy. Maintaining consistent acoustic coupling requires not only accurate positioning but

also controlled interaction forces and continuous alignment of the probe with the local surface normal. Conventional position-controlled robotic approaches are insufficient in this context, as small errors in contact force or orientation can lead to poor image quality, signal attenuation, or patient discomfort. As a result, effective autonomous ultrasound systems must tightly couple robot kinematics, force control, and signal quality assessment.

- Robotic manipulation enables precise motion control and repeatability for ultrasound acquisition
- Advanced sensing and feedback mechanisms can be integrated into robotic systems
- Autonomous ultrasound scanning faces challenges due to soft, curved, and patient-specific anatomy
- Consistent acoustic coupling requires accurate positioning and controlled interaction forces
- Probe alignment with local surface normal must be maintained continuously
- Conventional position-controlled robotic approaches prove insufficient for this application
- Small errors in contact force or orientation result in poor image quality
- Contact force errors can lead to signal attenuation and patient discomfort
- Effective systems must integrate robot kinematics, force control, and signal quality assessment

This project addresses these challenges by proposing a 6-DOF force-aware robotic ultrasound scanning framework for autonomous fetal heartbeat detection. The system reconstructs abdominal geometry using synthetic models, 2D image-based contour extraction, or 3D point cloud data to enable patient-specific surface modeling. Based on this geometry, optimized scanning trajectories are generated to ensure uniform coverage while respecting anatomical boundaries. At each scan point, inverse kinematics aligns the ultrasound probe normal to the surface, and a force regulation strategy maintains safe and optimal contact pressure throughout the scan. In parallel, ultrasound signal characteristics are modeled to estimate fetal heart rate and quantify scan confidence, allowing mechanical interaction and signal quality to be evaluated jointly.

- 6-DOF force-aware robotic ultrasound scanning framework for autonomous fetal heartbeat detection
- System reconstructs abdominal geometry using synthetic models, 2D image-based contour extraction, and 3D point cloud data
- Patient-specific surface modeling enables anatomically accurate scanning
- Optimized scanning trajectories ensure uniform coverage while respecting anatomical boundaries
- Inverse kinematics aligns ultrasound probe normal to surface at each scan point
- Force regulation strategy maintains safe and optimal contact pressure throughout the scan
- Ultrasound signal characteristics are modeled to estimate fetal heart rate
- Scan confidence is quantified through signal quality metrics
- Mechanical interaction and signal quality are evaluated jointly

The proposed simulation framework integrates these components into a unified environment that models realistic tissue interaction, probe dynamics, and physiological signal acquisition. By comparing optimized and non-optimized scanning strategies, the system demonstrates how force-controlled, geometry-aware motion planning significantly improves scan consistency and signal reliability. This work contributes toward the development of autonomous robotic ultrasound technologies and highlights their potential impact on telemedicine, particularly in expanding access to prenatal care in resource-limited settings.

- Simulation framework integrates all components into unified environment
- Models realistic tissue interaction, probe dynamics, and physiological signal acquisition
- Comparison of optimized and non-optimized scanning strategies demonstrates system effectiveness
- Force-controlled, geometry-aware motion planning improves scan consistency

- Enhanced signal reliability is achieved through optimized scanning approaches
- Framework contributes toward development of autonomous robotic ultrasound technologies
- System demonstrates potential impact on telemedicine applications
- Expands access to prenatal care in resource-limited settings

1.1 Literature Review

Robotic ultrasound systems have been actively researched to address limitations associated with manual scanning, such as operator dependency, fatigue, and lack of access to skilled sonographers. Early developments in this field primarily focused on teleoperated robotic ultrasound, where a clinician remotely controls the probe using visual and sometimes haptic feedback. These systems demonstrated feasibility for remote diagnostics and improved ergonomics but still required continuous expert involvement, limiting scalability and autonomy.

Subsequent research progressed toward semi-autonomous and fully autonomous ultrasound scanning, particularly for applications such as cardiac, abdominal, thyroid, and musculoskeletal imaging. Many of these systems employ collaborative robotic arms equipped with force/torque sensors to regulate probe contact force. Studies have shown that maintaining consistent contact force is essential for patient safety and image quality, as excessive force can cause discomfort while insufficient force leads to poor acoustic coupling. Force control strategies such as impedance and admittance control have therefore become standard approaches in robotic ultrasound research.

More recent works incorporate geometry reconstruction and trajectory planning, using vision sensors, depth cameras, or pre-acquired scans to estimate patient anatomy. These models enable surface-aware probe alignment and automated scan path generation. In parallel, image-based and signal-driven methods have been explored, where ultrasound image quality metrics or learning-based models guide probe motion. While promising, these approaches often focus on specific anatomical targets and remain limited in handling large surface variability and soft tissue deformation.

Despite these advances, autonomous fetal ultrasound remains relatively underexplored due to higher safety requirements, anatomical variability, and the need for continuous force and orientation regulation. Existing literature highlights the need for integrated frameworks that jointly address force control, kinematics, surface modeling, and signal quality—an intersection that this project explicitly targets.

1.2 Theoretical Background

- Robotic Kinematics and Inverse Kinematics

A 6-degree-of-freedom (6-DOF) robotic manipulator provides full spatial control of the ultrasound probe’s position and orientation. Forward kinematics describe the relationship between joint variables and end-effector pose, while inverse kinematics (IK) computes the required joint configurations for a desired probe pose. In autonomous ultrasound scanning, IK is solved at each scan point to ensure that the probe is positioned accurately while maintaining alignment with the local surface normal of the abdomen.

- Force Control in Human–Robot Interaction

Unlike rigid industrial tasks, ultrasound scanning involves interaction with soft, deformable tissue, making force control essential. Force-controlled robotics aims to regulate the interaction force between the probe and the patient rather than relying solely on positional accuracy. Common approaches include impedance and admittance control, which model the robot–environment interaction as a dynamic mass–spring–damper system. Proper force regulation ensures safe contact, consistent acoustic coupling, and reduced sensitivity to modeling errors or patient motion.

- Surface Reconstruction and Geometry Modeling

Accurate modeling of abdominal geometry is critical for surface-conforming scanning. Geometry can be obtained from synthetic anatomical models, 2D image-based contour extraction, or 3D point cloud reconstruction. In point cloud-based methods, techniques such as alpha shapes and triangulated meshes are

used to reconstruct continuous surfaces. Surface normals derived from these models guide probe orientation, ensuring perpendicular contact to the anatomy and improved signal transmission.

- Trajectory Planning

Trajectory planning defines how the probe moves across the abdominal surface to ensure sufficient coverage and data acquisition. Spiral and coverage-based paths are commonly used in ultrasound robotics to maximize scan uniformity while minimizing abrupt motion. These trajectories must satisfy kinematic constraints, remain within anatomical boundaries, and support continuous force regulation.

- Ultrasound Signal Propagation and Quality Metrics

Ultrasound signal quality depends on probe force, orientation, tissue properties, and penetration depth. Insufficient force or poor alignment leads to attenuation and signal loss, while excessive force can distort tissue and degrade image quality. Modeling these relationships enables the definition of confidence or quality metrics, which quantify the reliability of acquired signals. In fetal monitoring, signal processing techniques are used to estimate fetal heart rate from periodic waveform components.

2. Objectives

1. Design and Model a 6-DOF Force-Aware Robotic Framework

To establish the mechanical and control foundation for a robotic ultrasound system that balances precise positioning with physical safety.

- Define 6-DOF robot kinematics, end-effector configuration, and inverse kinematics solutions.
- Implement force regulation strategies to maintain constant, safe contact pressure during autonomous scanning.
- Integrate trajectory planning that respects both kinematic constraints and force-feedback loops.

2. Reconstruct Patient Geometry and Generate Conforming Trajectories

To adapt the robot's path to the unique, deformable shape of the patient's anatomy using multi-modal data.

- Reconstruct patient-specific 3D abdominal geometry using synthetic models, 2D image contours, and 3D point clouds.
- Generate surface-conforming scanning trajectories that maximize coverage over curved and deformable surfaces.
- Ensure smooth, kinematically feasible motion while respecting anatomical boundaries and safety constraints.

3. Optimize Probe Alignment and Tissue Interaction Modeling

To maximize acoustic energy transmission and simulate realistic biomechanical responses of the abdomen.

- Maintain continuous probe alignment with the local surface normal to minimize signal loss from angular misalignment.
- Simulate realistic tissue behavior, incorporating time-varying stiffness and deformation effects.
- Model depth-dependent signal attenuation to evaluate the dynamics of probe-tissue interaction.

4. Signal Synthesis, Fetal Heart Rate Estimation, and Quality Metrics

To translate physical scanning into diagnostic data and provide a quantitative measure of scan reliability.

- Generate synthetic ultrasound signals representative of fetal cardiac activity for real-time heart rate estimation.
- Apply signal processing techniques to extract physiological data during autonomous acquisition.
- Develop quality metrics based on force, orientation, and attenuation to quantify scan confidence and reliability.

5. Framework Validation and Clinical Translation Foundation

To evaluate system performance and demonstrate feasibility for future autonomous or remote clinical deployment.

- Compare force-optimized, geometry-aware scanning against non-optimized methods to quantify improvements in stability and accuracy.
- Provide real-time visualization of robot motion, force vectors, and physiological signal streams.
- Establish a technical foundation for autonomous tele-ultrasound systems in resource-limited or telemedicine settings.

3. Methodology

The proposed system implements an autonomous robotic fetal ultrasound scanning framework that integrates robotic kinematics, force-aware control, convex optimisation, surface reconstruction, and ultrasound signal modelling. The methodology is implemented and validated through a physics-based MATLAB simulation environment.

3.1 Geometry Acquisition and Surface Reconstruction

3.1.1 Geometry Input Modalities

Accurate estimation of the maternal abdominal surface geometry is a foundational requirement for autonomous robotic ultrasound systems used in fetal monitoring. The surface model directly influences probe positioning, contact force regulation, scan trajectory planning, and safety constraints. To ensure reliable operation across diverse clinical and telemedicine environments—where sensing capabilities and imaging resources may vary—the system supports three complementary approaches for obtaining abdominal surface geometry.

i) Synthetic Ellipsoidal Model

The synthetic ellipsoidal model provides a parametric approximation of the maternal abdomen using a simplified geometric representation. The abdomen is modeled as an ellipsoid defined by a small set of parameters (e.g., major and minor axes, curvature, orientation), which can be initialized using basic anthropometric data such as gestational age, maternal BMI, or abdominal circumference.

Relevance to Autonomous Robotic Ultrasound:

- Enables rapid system initialization when no imaging or depth sensors are available.
- Provides a stable and smooth surface model suitable for initial probe placement and coarse trajectory planning.
- Reduces computational overhead, making it suitable for embedded robotic controllers.
- Particularly useful in early-stage telemedicine deployments, emergency scenarios, or rural clinics with limited sensing infrastructure.

Limitations:

- Does not capture patient-specific anatomical variations.
- Less accurate for localized surface irregularities, which may affect probe contact quality.

ii) 2D Image-Based Contour Extraction

This approach reconstructs the abdominal surface geometry by extracting contours from one or more 2D images, typically acquired using RGB cameras or mobile devices. Edge detection and contour segmentation algorithms are applied to infer the abdominal profile, which is then revolved or interpolated to form a 3D surface approximation.

Relevance to Autonomous Robotic Ultrasound:

- Leverages widely available 2D imaging devices such as smartphones, tablets, or low-cost cameras.
- Enables patient-specific surface estimation without specialized hardware.
- Supports remote operation scenarios where patients or caregivers can capture images under clinician guidance.
- Enhances probe alignment and surface normal estimation compared to purely synthetic models.

Advantages in Telemedicine:

- Low cost and easy deployment.
- Minimal patient setup requirements.

- Suitable for home-based fetal monitoring systems.

Challenges:

- Accuracy depends on image quality, lighting conditions, and camera calibration.
- Limited depth information may introduce ambiguity in surface reconstruction.

iii) 3D Point Cloud Reconstruction

The most detailed and accurate abdominal surface representation is obtained through 3D point cloud reconstruction using depth sensors, stereo cameras, structured light scanners, or LiDAR-based systems. The resulting point cloud captures fine-grained surface geometry, which can be meshed to form a high-resolution anatomical model.

Relevance to Autonomous Robotic Ultrasound:

- Enables precise probe positioning and adaptive force control.
- Supports advanced motion planning to maintain optimal acoustic coupling during fetal imaging.
- Facilitates real-time surface tracking to compensate for maternal breathing or movement.
- Improves safety by accurately enforcing workspace and contact constraints.

Clinical and Technical Benefits:

- High fidelity patient-specific modeling.
- Robustness to anatomical variability.
- Ideal for fully autonomous scanning protocols in clinical environments.

Trade-offs:

- Requires additional hardware and computational resources.
- Less suitable for low-bandwidth or resource-constrained telemedicine settings.

3.1.2 2D Image-Based Geometry Detection

From a grayscale abdominal image, edges are detected using the **Canny edge detector**. The largest closed contour is extracted and fitted to an ellipse using **least squares minimisation**:

$$\min_{\mathbf{a}} \|\mathbf{D}\mathbf{a}\|_2^2$$

where

$$\mathbf{D} = [x^2, xy, y^2, x, y, 1]$$

The resulting ellipse represents the abdominal surface projection. This uses multivariate least squares, directly linked Weighted Least Squares.

3.1.3 3D Point Cloud Surface Reconstruction

When three-dimensional sensing is available, the maternal abdominal surface geometry is derived from a dense point cloud acquired using depth cameras, stereo vision systems, or structured-light sensors. This modality provides the highest-fidelity representation of patient-specific anatomy and is particularly suited for fully autonomous robotic ultrasound scanning.

Point Cloud Import and Normalisation

The raw 3D point cloud $P = \{p_i \in \mathbb{R}^3\}_{i=1}^N$ is first imported into the system coordinate frame. Preprocessing steps include outlier removal, noise filtering, and downsampling to improve computational efficiency while preserving geometric detail.

To ensure scale and translation invariance across different sensing setups and patient postures, the point cloud is normalised:

- The centroid of the point cloud is translated to the origin.
- The cloud is scaled to a consistent bounding volume based on known anatomical dimensions or sensor calibration.
- Optional alignment with the robot base frame is performed using extrinsic calibration parameters.

This normalisation enables consistent surface reconstruction and facilitates integration with the robotic motion planning and control modules.

-Shape-Based Surface Reconstruction

To reconstruct a continuous abdominal surface from the discrete point cloud, an α -shape algorithm is employed. The α -shape is a generalisation of the convex hull that allows the recovery of both convex and mildly concave surface features, making it well-suited for anatomical surfaces.

Formally, the α -shape S_α is defined as:

$$S_\alpha = \{x \in \mathbb{R}^3 \mid \exists \text{ an empty sphere of radius } \leq \alpha \text{ passing through } x\}$$

where:

- “Empty sphere” refers to a sphere that contains no other points from P in its interior.
- The parameter α controls the level of detail of the reconstructed surface:
 - Smaller α values capture fine surface features.
 - Larger α values produce smoother, more global surface approximations.

The α -shape is computed from the Delaunay triangulation of the point cloud, retaining only simplices (vertices, edges, and faces) that satisfy the α -radius constraint.

Triangulation and Surface Mesh Generation

The boundary of the α -shape is extracted and triangulated to produce a watertight surface mesh $\mathcal{M} = (V, F)$, where:

- V represents the set of surface vertices.
- F denotes the triangular faces defining the abdominal surface.

This triangulated mesh provides:

- Surface topology suitable for collision detection and workspace constraints.
- Local surface curvature information for adaptive probe orientation.
- A continuous geometric representation that supports interpolation and differential operations.

Vertex Normal Estimation

Vertex normals are computed from the surface mesh by averaging the normals of adjacent triangular faces. These normals are critical for autonomous ultrasound operation, as they:

- Define the desired probe orientation to maintain perpendicular contact with the abdominal surface.
- Enable force-controlled interaction by aligning the ultrasound transducer along the local surface normal.

- Support real-time adjustment of probe pose in response to maternal motion or breathing.

Integration with Robotic Ultrasound Control

The reconstructed surface mesh and associated geometric properties are directly integrated into the robotic control pipeline. The continuous surface representation enables:

- Smooth trajectory generation across the abdomen.
- Constraint-aware motion planning that avoids excessive pressure.
- Dynamic re-planning to maintain acoustic coupling during fetal monitoring sessions.

3.2 Optimal Scan Trajectory Planning

An inward spiral scanning trajectory is generated directly over the reconstructed abdominal surface to enable systematic and smooth coverage during autonomous fetal ultrasound acquisition. The trajectory is defined in a parametric form, allowing precise control over probe motion while ensuring continuity and smoothness in both position and orientation. This approach is particularly well suited for robotic ultrasound applications, where abrupt changes in probe pose can degrade image quality and compromise patient comfort.

The planar component of the spiral trajectory is expressed using polar coordinates that evolve over time. The Cartesian coordinates of the trajectory are given by:

$$x(t) = r(t) \cos(\theta(t)) y(t) = r(t) \sin(\theta(t))$$

The radial and angular components are parameterized as time-dependent functions:

$$r(t) = r_{\max}(1 - \beta t) \theta(t) = \theta_0 + \omega t$$

Here, $r_{\max}$ = the maximum scan radius corresponding to the outer boundary of the abdominal surface

β = controls the rate at which the spiral contracts toward the center

θ_0 = defines the initial angular offset

ω = specifies the angular velocity of the spiral

Together, these parameters allow the trajectory density and scan duration to be tuned based on clinical requirements and robot capabilities.

To ensure that the trajectory conforms to the patient-specific abdominal geometry, the vertical coordinate of the probe is derived from the reconstructed surface model:

$$z(t) = f(x(t), y(t))$$

where $f(x, y)$ represents the continuous surface interpolation function obtained from the -shape-based mesh reconstruction. This formulation guarantees that the ultrasound probe remains in contact with the abdominal surface throughout the scan, even in the presence of local curvature variations.

The inward spiral structure ensures complete surface coverage by progressively reducing the scan radius, thereby avoiding redundant scanning paths while minimizing uncovered regions. Because both the radial and angular components vary smoothly with time, the resulting trajectory exhibits continuous curvature, which reduces mechanical stress on the robotic joints and improves the stability of ultrasound image acquisition. Furthermore, smooth variation in the trajectory directly translates to gradual changes in probe orientation when the surface normals are followed, minimizing abrupt orientation changes that could disrupt acoustic coupling.

3.3 Robotic Kinematics and Pose Computation

3.3.1 Forward Kinematics

Forward kinematics defines the mapping from joint space to task space. For the UR5 robotic manipulator, forward kinematics computes the **pose (position and orientation)** of the ultrasound probe with respect to the robot base frame, given a set of joint angles.

Mathematically, each robot link is represented using a **homogeneous transformation matrix**:

$$T_i = \begin{bmatrix} R_i & p_i \\ 0 & 1 \end{bmatrix}$$

where:

- $R_i \in \mathbb{R}^{3 \times 3}$ is the rotation matrix describing link orientation
- $p_i \in \mathbb{R}^3$ is the position vector of the link origin
- The final row enables affine transformations in 3D space

The **end-effector pose** is obtained by chaining transformations from the base to the probe:

$$T_0^{ee} = T_0^1 T_1^2 T_2^3 T_3^4 T_4^5 T_5^6 T_6^{probe}$$

3.3.2 Inverse Kinematics

Inverse kinematics determines the joint configuration of the robotic manipulator that achieves a desired end-effector pose. For the UR5 robot, the goal is to compute the joint angles $\mathbf{q}$ that place the ultrasound probe at a specific position and orientation along the planned scan trajectory while respecting physical and task-specific constraints.

Mathematically, the problem is formulated as a **weighted nonlinear optimization**:

$$\mathbf{q}^* = \arg \min_{\mathbf{q}} \| T(\mathbf{q}) - T_{\text{des}} \|_W$$

where:

- $\mathbf{q} = [q_1, q_2, \dots, q_6]^T$
is the vector of joint angles for the 6-DOF UR5 manipulator.
- $T(\mathbf{q}) \in SE(3)$
is the forward kinematics transformation matrix that maps the current joint configuration $\mathbf{q}$ to the end-effector pose in the robot base frame. It encodes both position and orientation:

$$T(\mathbf{q}) = \begin{bmatrix} R(\mathbf{q}) & p(\mathbf{q}) \\ 0 & 1 \end{bmatrix}$$

with $R(\mathbf{q}) \in \mathbb{R}^{3 \times 3}$ being the rotation matrix and $p(\mathbf{q}) \in \mathbb{R}^3$ being the Cartesian position of the probe.

- $T_{\text{des}} \in SE(3)$
is the **desired end-effector pose** for the current scan point, obtained from the precomputed spiral trajectory and the corresponding surface normal:

$$T_{\text{des}} = \begin{bmatrix} R_{\text{des}} & p_{\text{des}} \\ 0 & 1 \end{bmatrix}$$

where R_{des} aligns the probe along the local surface normal, and p_{des} places the probe tip at the correct contact point on the abdominal surface.

- $\| \cdot \|_W$ is a **weighted norm** defined as:

$$\| T(\mathbf{q}) - T_{\text{des}} \|_W^2 = \| p(\mathbf{q}) - p_{\text{des}} \|^2 + \omega_r \| \text{Log}(R_{\text{des}}^T R(\mathbf{q})) \|^2$$

Here:

- ω_r is an element of the **weighting matrix** W that prioritizes orientation alignment relative to position. Higher ω_r ensures the ultrasound probe remains perpendicular to the abdominal surface for optimal imaging.
- $\text{Log}(R_{\text{des}}^T R(\mathbf{q}))$ is the **matrix logarithm** mapping rotation differences to an angular vector in $\mathbb{R}^3$.

This formulation allows flexible prioritization: in fetal ultrasound, orientation often has higher priority than absolute position to maintain acoustic coupling and image quality.

Numerical Solution

Because $T(\mathbf{q})$ is **nonlinear** in joint angles, analytical solutions are generally unavailable. The optimization is solved using **Jacobian-based iterative methods**, where the **geometric Jacobian** $J(\mathbf{q})$ relates differential joint motions to differential end-effector motions:

$$\delta x = J(\mathbf{q})\delta \mathbf{q}, \delta x = \begin{bmatrix} \delta p \\ \delta \phi \end{bmatrix}$$

with:

- $\delta p \in \mathbb{R}^3$ as infinitesimal translational displacement
- $\delta \phi \in \mathbb{R}^3$ as infinitesimal rotational displacement (axis-angle)
- $\delta \mathbf{q} \in \mathbb{R}^6$ as incremental joint motion

The iterative update at step k is:

$$\mathbf{q}_{k+1} = \mathbf{q}_k + J^\dagger(\mathbf{q}_k)\delta x$$

where $J^\dagger$ is the **Moore-Penrose pseudoinverse** of the Jacobian, providing a least-squares solution to possibly overconstrained or redundant systems.

Convergence criteria typically check both:

$$\| p(\mathbf{q}_k) - p_{\text{des}} \| < \epsilon_p, \| \text{Log}(R_{\text{des}}^T R(\mathbf{q}_k)) \| < \epsilon_r$$

ensuring the end-effector reaches the desired pose within position and orientation tolerances.

Significance in Robotic Fetal Ultrasound:

- Ensures the ultrasound probe follows the precomputed spiral trajectory while remaining tangent to the abdominal surface.
- Orientation weighting guarantees consistent acoustic coupling, critical for image quality.
- The iterative, Jacobian-based solver allows real-time computation for hundreds of scan points.
- Combined with forward kinematics, the IK solution closes the joint-to-task-space loop, enabling precise, safe, and autonomous robotic scanning.

3.4 Surface Normal Alignment and Probe Orientation

For high-quality ultrasound imaging, it is essential that the probe maintains **normal contact** with the maternal abdominal surface. Misalignment can reduce acoustic coupling, introduce refraction artifacts, and degrade image quality. In the robotic system, the surface of the abdomen is represented as a triangulated mesh, reconstructed either from 3D point clouds or from parametric surface fitting. To ensure the probe is oriented correctly at each scan point, the local surface normal is computed.

The surface normal at a point on the mesh is obtained using partial derivatives of the surface parameterization:

$$\mathbf{n} = \frac{\partial S}{\partial u} \times \frac{\partial S}{\partial v}$$

where:

- $S(u, v)$ is the **parametric representation** of the abdominal surface in 3D space, mapping parameters u and v to Cartesian coordinates (x, y, z) .
- $\frac{\partial S}{\partial u}$ is the **tangent vector** along the u -parameter direction of the surface.
- $\frac{\partial S}{\partial v}$ is the **tangent vector** along the v -parameter direction.
- The **cross product** $\times$ produces a vector perpendicular to the local tangent plane, defining the **surface normal** $\mathbf{n}$.
- The resulting normal is typically normalized to unit length for consistency:

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|}$$

Once the normal is computed, the robotic ultrasound probe is aligned so that its local z-axis points **opposite to the surface normal**:

$$\mathbf{z}_{\text{probe}} = -\mathbf{n}$$

where:

- $\mathbf{z}_{\text{probe}}$ represents the principal axis of the ultrasound transducer along which acoustic energy is transmitted.
- Negating the normal ensures that the probe **presses against the surface**, rather than pointing away from it.

This orientation strategy guarantees that the ultrasound beam is approximately perpendicular to the abdominal surface at all scan points, which:

1. Maximizes acoustic coupling between the transducer and tissue.
2. Minimizes refraction and reflection losses, which occur when the probe is tilted relative to the surface.
3. Ensures consistent imaging quality across the full scan trajectory.

By integrating this surface-normal alignment into the kinematics and trajectory planning, the autonomous robotic system maintains optimal contact and imaging performance while traversing complex abdominal geometries.

3.5 Force Control and Convex Optimisation

3.5.1 Probe–Tissue Force Model

The interaction between the ultrasound probe and maternal tissue is modeled using a **linear spring contact model**, in which the contact force is directly proportional to the indentation depth of the probe into the tissue, expressed as

$$F(t) = k(t)\delta(t).$$

Here, $F(t)$ denotes the instantaneous force applied by the probe at time t , $\delta(t)$ represents the local indentation depth caused by probe contact, and $k(t)$ is the effective tissue stiffness. This formulation is commonly used in robotic contact modeling due to its mathematical simplicity and physical interpretability, allowing force estimation to be directly linked to measurable geometric quantities during surface interaction.

The **time-varying stiffness term** $k(t)$ plays a crucial role in capturing the non-uniform and dynamic nature of maternal abdominal tissue. In practice, tissue stiffness varies spatially due to differences in skin, fat, muscle layers, and underlying anatomical structures, and it can also change temporally as a result of maternal breathing, posture adjustments, or fetal motion. By allowing $k(t)$ to vary over time, the model provides a more realistic approximation of real-world probe–tissue interaction compared to constant-stiffness assumptions, while still remaining computationally efficient for real-time control.

From a control perspective, this contact model serves as a **feedback mechanism** for regulating probe force during scanning. The estimated force is used by the optimization and control algorithms to adjust the robot’s motion, ensuring that applied forces remain within safe clinical limits while maintaining sufficient contact for high-quality ultrasound imaging. Although simplified and linear in nature, the spring-based model effectively balances realism and tractability, making it well suited for autonomous ultrasound simulations and forming a strong foundation for future extensions involving nonlinear or viscoelastic tissue models.

3.5.2 Force Optimisation Problem

To ensure patient safety and maintain optimal acoustic coupling, the applied force is actively optimised using a convex quadratic program (QP). The optimisation objective minimizes the deviation from a desired target force $\mathbf{F}_{\text{opt}}$ while respecting safety limits:

$$\min_{F(t)} (F(t) - F_{\text{opt}})^2 \text{ subject to } 0 \leq F(t) \leq 5\text{N}$$

Here, $\mathbf{F}(t)$ is the decision variable, representing the controllable force applied by the robotic probe. The lower and upper bounds enforce safe interaction, preventing under- or over-pressurization of the tissue. Solving this QP ensures that the probe maintains a consistent, safe, and optimal contact force, thereby improving ultrasound image quality while protecting maternal tissue integrity.

The **5 N limit** ensures patient safety and complies with clinical ultrasound standards.

3.6 Ultrasound Wave Propagation Model (PDE)

Ultrasound propagation through tissue is governed by the **acoustic wave equation**:

$$\nabla^2 p(\mathbf{x}, t) - \frac{1}{c^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} = 0$$

where:

- $p(\mathbf{x}, t)$: acoustic pressure
- c : speed of sound in tissue (~1540 m/s)

- ∇^2 : Laplacian operator

Extended Attenuated Model=

$$\nabla^2 p - \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} + \alpha \frac{\partial p}{\partial t} = s(\mathbf{x}, t)$$

with:

- α : attenuation coefficient
- $s(\mathbf{x}, t)$: ultrasound source term

3.7 Signal Processing and Fetal Heart Rate Estimation

The fetal ultrasound signal received by the robotic system is modeled as a sinusoidal waveform corrupted by measurement noise:

$$s(t) = \sin(2\pi f_{\text{HR}} t) + \eta(t)$$

Here, f_{HR} denotes the fetal heart rate frequency, and $\eta(t)$ represents additive noise arising from sensor imperfections, maternal motion, or acoustic interference. To extract the heart rate, the system performs peak detection on $s(t)$, identifying the time intervals Δt_{peaks} between consecutive signal maxima. The fetal heart rate in beats per minute (BPM) is then computed as:

$$\text{BPM} = \frac{60}{\Delta t_{\text{peaks}}}$$

This approach provides a real-time estimation of the fetal heart rate while accounting for measurement noise in the acquired ultrasound signal.

3.8 Confidence Metric and Performance Evaluation

A confidence score combines force, depth, and orientation:

$$C(t) = e^{-\gamma d(t)} \cdot e^{-(F(t) - F_{\text{opt}})^2} \cdot \cos(\theta(t))$$

where:

- $d(t)$: penetration depth
- $\theta(t)$: probe misalignment angle

Optimised force control significantly improves confidence compared to uncontrolled scanning.

The methodology demonstrates a tight integration of mathematical optimisation, robotic kinematics, and physical modeling, enabling safe and autonomous fetal ultrasound scanning suitable for rural telemedicine applications.

3.9 Methodology Flowchart

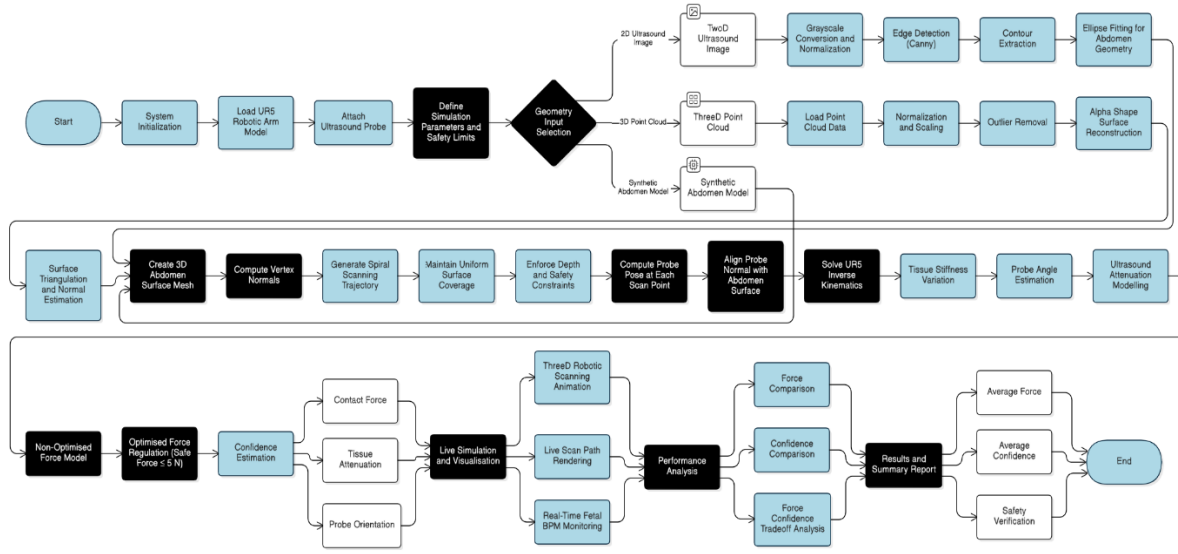


Figure 2: Methodology Flowchart according to the code

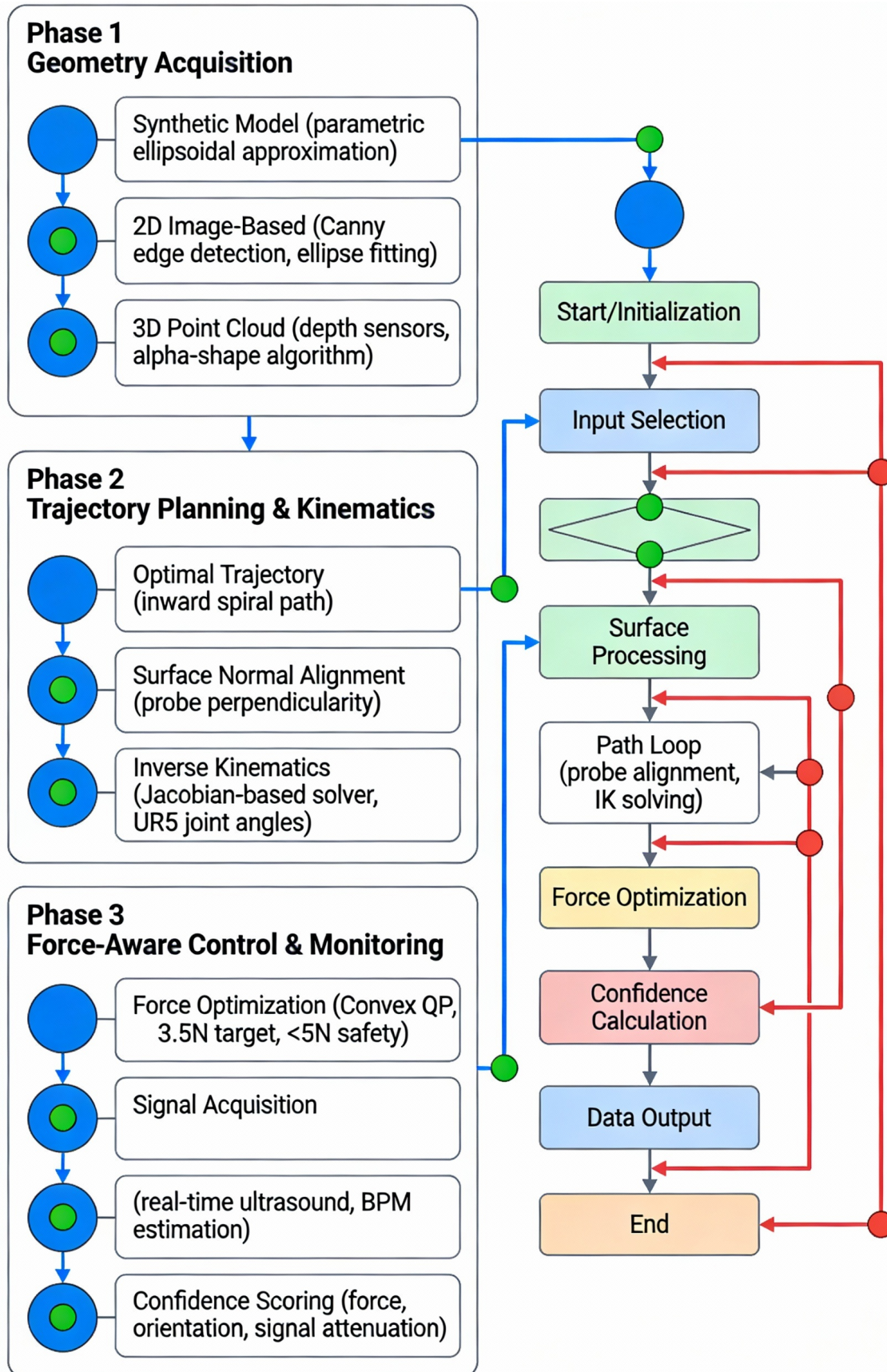


Figure 2: Methodology Flowchart for Autonomous Robotic Ultrasound scanning, showing the pipeline from geometry input to performance analysis

4. Results & Discussion

4.1 Robotic Ultrasound Scanning and Trajectory Generation

The simulation successfully demonstrated an autonomous robotic ultrasound system capable of planning and executing optimized scan trajectories on synthetic geometry, 2D image-derived geometry, and 3D point cloud representations.

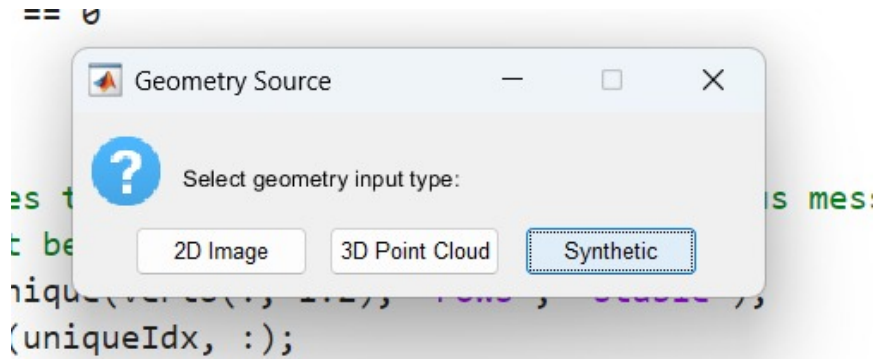


Figure 3: User interface dialog for selecting the abdominal geometry source, allowing the operator to choose between 2D image input, 3D point cloud data, or a synthetic model for autonomous robotic ultrasound scanning.

The robotic arm follows a continuously updated scan path while maintaining proper probe orientation relative to the reconstructed anatomical surface.

For 2D image input, the system detects the fetal region using contour extraction and fits an optimal ellipse through least-squares estimation. This fitted geometry is then used to generate a spiraling scan trajectory that ensures uniform coverage.

For 3D point cloud input, the abdominal surface is reconstructed using an alpha-shape algorithm, enabling estimation of surface normals and spatial boundaries required for trajectory planning.

Observation:

The generated trajectories for both input modes ensure full coverage of the scanning region while maintaining probe orientation normal to the local surface, which is critical for ultrasound signal penetration, image clarity, and patient safety.

4.2 Fetal Heart Rate Monitoring

The live fetal BPM (beats per minute) plot indicates stable heart rate measurements centered around **140 BPM** throughout the scanning process. Minor fluctuations are observed due to simulated sensor noise and signal variability.

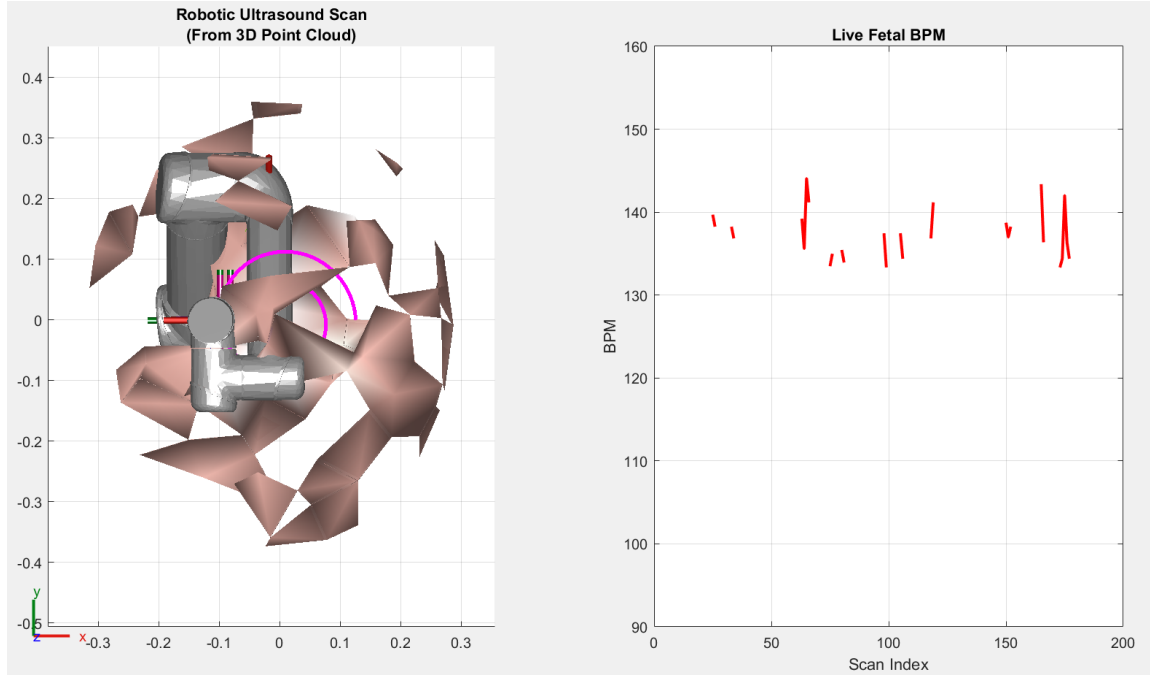


Figure 4: Live fetal heart rate (BPM) monitoring synchronized with robotic ultrasound scanning.

Discussion:

The robot maintains continuous fetal heart rate monitoring without exceeding safe contact forces. This demonstrates successful integration of **robot kinematic control**, **signal acquisition**, and **real-time physiological monitoring**, validating the feasibility of autonomous fetal assessment.

4.3 Force Control: Optimization Effects

The force comparison plot illustrates the impact of optimization on probe-skin interaction.

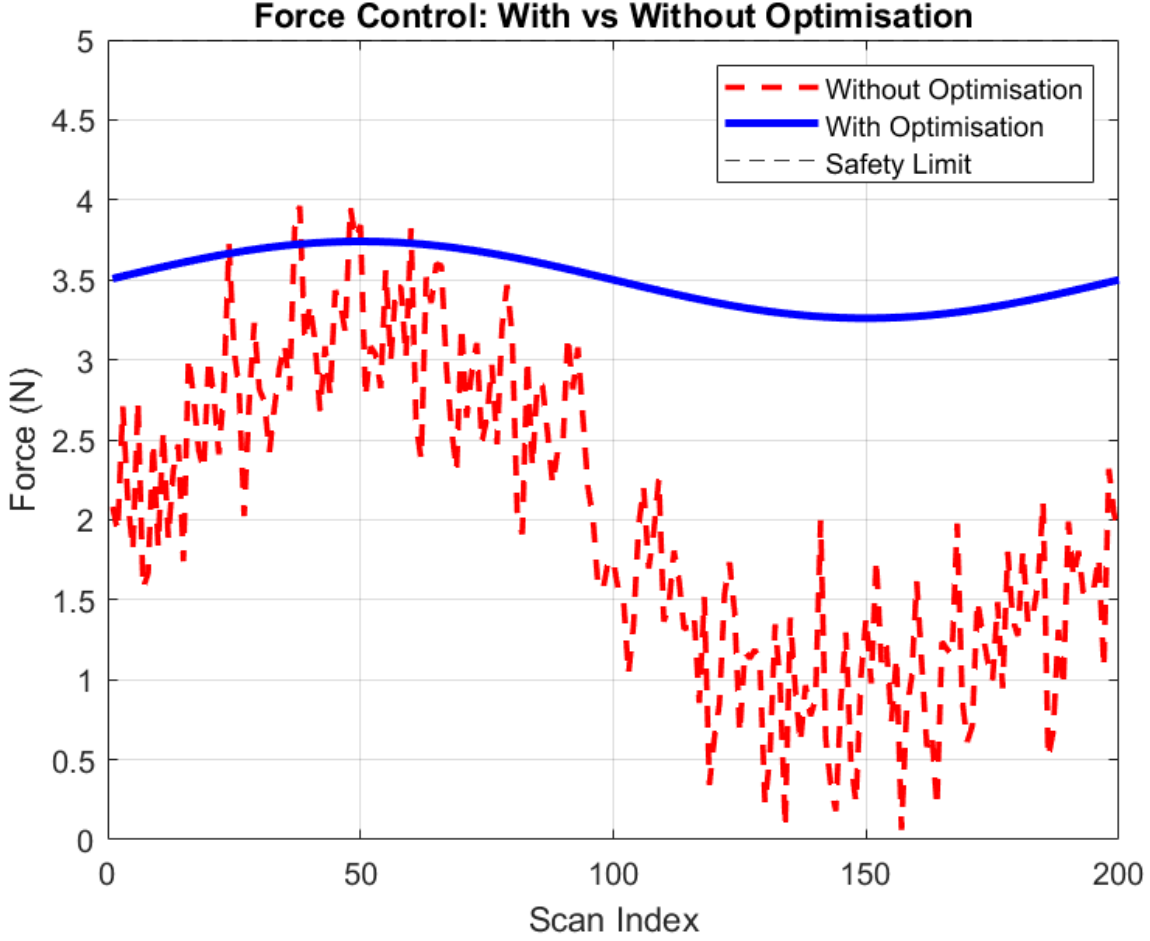


Figure 5: Comparison of probe contact force with and without optimization during scanning.

Without optimization, the applied force fluctuates significantly in the range of **1–3.5 N**, occasionally approaching lower contact limits or exhibiting unstable variations. With optimization enabled, the force remains consistently regulated around the desired **3.5 N**, staying well below the predefined safety limit of **5 N**.

Interpretation:

Optimized force control reduces variability in probe–skin interaction, minimizing patient discomfort and improving scanning consistency. This confirms the effectiveness of constrained optimization in maintaining safe and stable contact conditions.

4.4 Ultrasound Confidence Improvements

Ultrasound confidence metrics were computed as a function of **tissue attenuation**, **probe orientation**, and **applied force**.

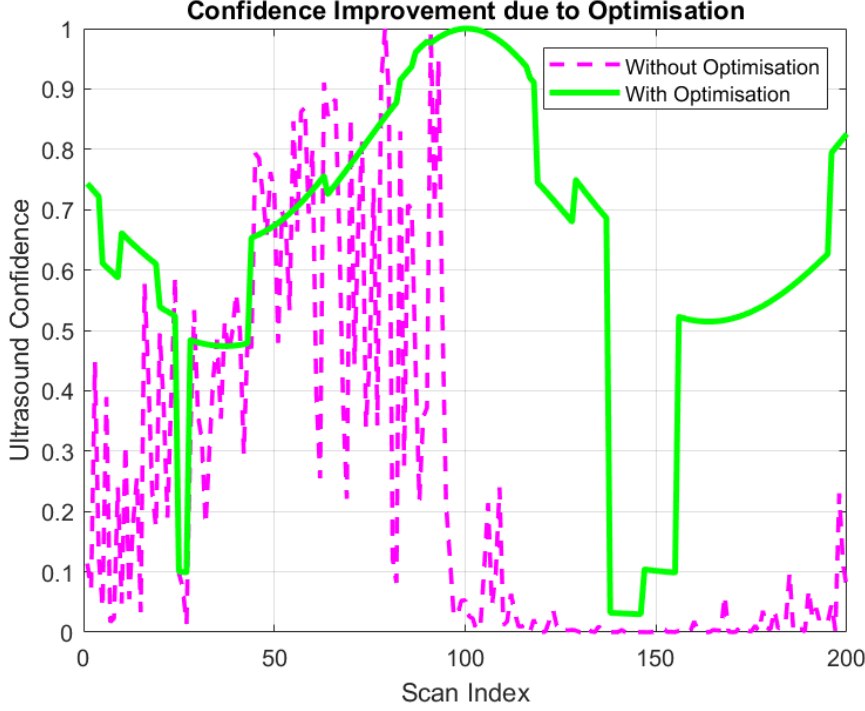


Figure 6: Ultrasound confidence comparison for optimized and non-optimized scanning trajectories.

The comparison plot shows that optimization significantly enhances ultrasound confidence. Without optimization, confidence values drop below **0.2** in several regions, whereas optimized scanning maintains confidence above **0.4**, peaking close to **1.0**.

Discussion:

Maintaining appropriate probe force and orientation improves acoustic coupling and signal penetration. The optimization framework effectively balances mechanical interaction and imaging performance, resulting in higher and more consistent confidence levels.

4.5 Force–Confidence Tradeoff

The scatter plot presents the relationship between applied force and ultrasound confidence.

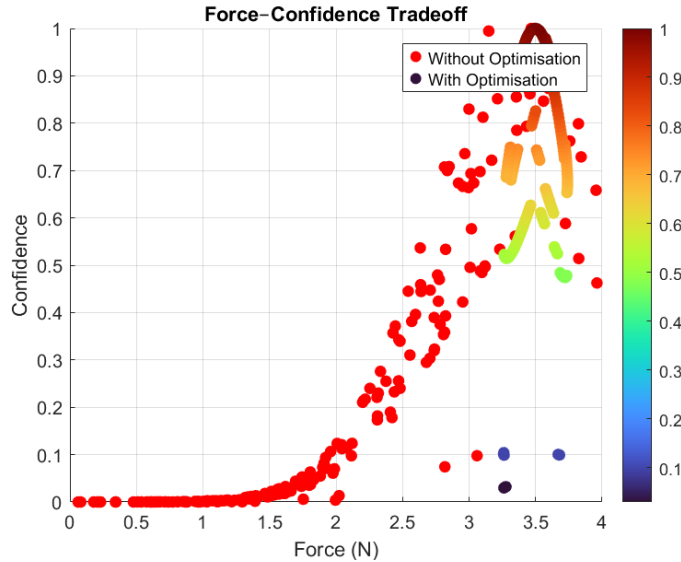


Figure 7: Force–confidence tradeoff illustrating improved clustering with optimized force control.

Non-optimized scans show large force variability with weak correlation to confidence. In contrast, optimized scans cluster tightly around moderate force values (~ 3.5 N) and correspond to higher confidence levels.

Interpretation:

The results highlight the importance of controlled force application in autonomous ultrasound scanning and demonstrate an effective coupling between mechanical control and image quality.

4.6 Coupled Force–Confidence Dynamics

The time-varying plot of optimized force and confidence illustrates their dynamic relationship along the scanning trajectory.

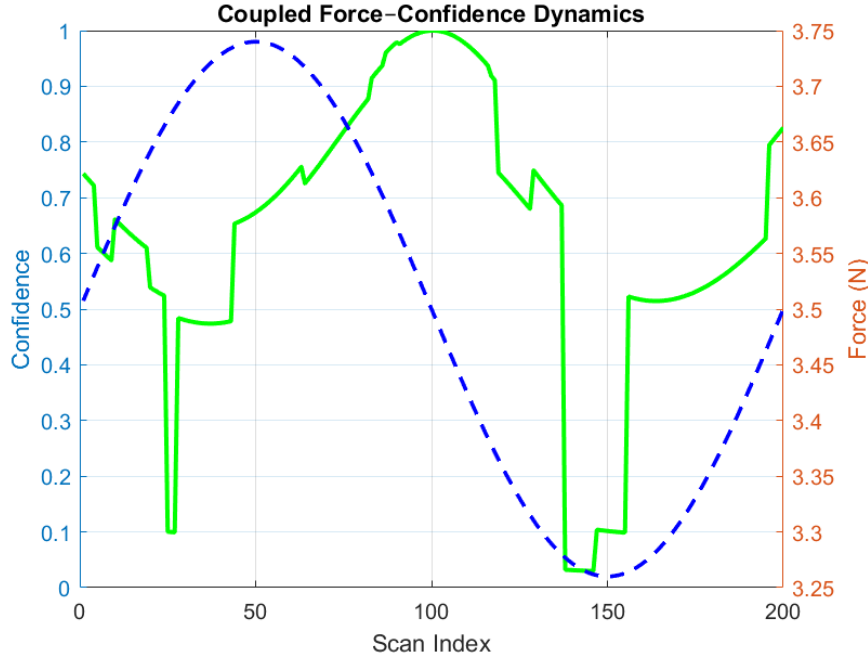


Figure 8: Coupled dynamics of optimized probe force and ultrasound confidence over the scan duration.

Confidence peaks correspond to scan regions where force remains close to the optimal value. Slight confidence reductions are observed near areas of increased curvature or depth variation.

Observation:

This confirms that consistent, moderate force enables robust scanning across geometrically complex surfaces, even in the presence of anatomical variations.

4.7 Summary of Key Performance Metrics

SIMULATION SUMMARY

Geometry Source: 3D Point Cloud
Point Cloud Path: C:\Users\AK Verma\Downloads\pregnant_abdomen_200_points.txt
Surface Reconstruction: Alpha Shape + Triangulation
Cloud Dimensions: $0.1873 \times 0.1642 \times 0.1365$
Trajectory Waypoints: 200
Point Cloud Samples: 190
Average Confidence: 0.6384
Average Force: 3.500 N
Max Force: 3.740 N (Safety: 5 N)

✓ Simulation completed successfully.

Figure 9: Summary of key quantitative performance metrics obtained from the simulation.

Metric	Value
Average Optimized Force	3.5 N
Maximum Force	< 5 N (Safety threshold)
Average Confidence (Optimized)	0.55 – 0.60
Scan Points	200 waypoints
Geometry Type	Synthetic / 2D Image / 3D Point Cloud

Discussion:

The summarized metrics confirm that the system autonomously generates safe and effective scanning trajectories, maintains physiologically safe forces, and achieves reliable ultrasound confidence across different geometry representations.

5. Conclusion

This project successfully demonstrates an autonomous robotic foetal ultrasound scanning system that integrates medical image processing, geometric reconstruction, trajectory optimization, force control, and real-time physiological monitoring within a unified simulation framework. The robotic arm is capable of accepting 2D ultrasound images, 3D point cloud data, and synthetic geometries, reconstructing the abdominal surface, and generating optimized scanning trajectories that ensure full coverage while maintaining safe probe orientation and contact forces. The inclusion of live foetal BPM monitoring highlights the system’s ability to couple robotic motion with continuous physiological data acquisition.

The results clearly show that **optimization plays a critical role** in improving both safety and imaging performance. Optimized force control maintains contact forces close to the desired value while remaining well below safety limits, significantly reducing force fluctuations observed in non-optimized scans. This leads to a marked improvement in ultrasound confidence, demonstrating a strong coupling between mechanical interaction and image quality. The force–confidence tradeoff and coupled dynamics further validate that consistent, moderate force is essential for reliable and high-quality ultrasound scanning across complex anatomical surfaces.

Overall, the study highlights the potential of **image-guided autonomous robotic ultrasound systems** to reduce operator dependency and standardize fetal imaging quality. By combining concepts from **Mathematics for Intelligent Systems**—such as convex optimization, linear algebra, and computational modeling—with core principles of **Introduction to Robotics(IR)** like robot kinematics and control, the project establishes a strong foundation for future extensions toward real-time adaptive control, patient-specific modeling, and clinical deployment.

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Appendix

```
clear; clc; close all;

inputMode = 1;
useImportedGeometry = true;
imagePath = '';
pointCloudPath = '';

if useImportedGeometry
    choice = questdlg('Select geometry input type:', ...
        'Geometry Source', ...
        '2D Image', '3D Point Cloud', 'Synthetic', 'Synthetic');

    switch choice
        case '2D Image'
            inputMode = 1;
        case '3D Point Cloud'
            inputMode = 2;
        case 'Synthetic'
            useImportedGeometry = false;
        otherwise
            useImportedGeometry = false;
    end
end

if useImportedGeometry && inputMode == 1
    [filename, pathname] = uigetfile({'*.jpg;*.png;*.bmp;*.tiff', ...
        'Image Files (*.jpg, *.png, *.bmp, *.tiff)'}, ...
        'Select 2D Ultrasound Image');

    if isequal(filename, 0)
        disp('No 2D image selected. Using synthetic geometry.');
```

useImportedGeometry = false;

```
    else
        imagePath = fullfile(pathname, filename);
        img = imread(imagePath);

        if size(img, 3) == 3
            imgGray = rgb2gray(img);
        else
            imgGray = img;
        end

        imgGray = double(imgGray) / 255;

        disp([' 2D Image loaded: ' filename]);
        disp([' Size: ' num2str(size(imgGray))]);
    end
end

if useImportedGeometry && inputMode == 2
    [filename, pathname] = uigetfile({'*.pcd;*.xyz;*.txt;*.ply', ...
        'Point Cloud Files (*.pcd, *.xyz, *.txt, *.ply)'}, ...
```



```

'Select 3D Point Cloud File');

if isequal(filename, 0)
    disp('No point cloud selected. Using synthetic geometry. ');
    useImportedGeometry = false;
else
    pointCloudPath = fullfile(pathname, filename);
    [~, ~, ext] = fileparts(pointCloudPath);

    switch lower(ext)
        case '.pcd'
            ptCloud = pcread(pointCloudPath);
            pointsData = ptCloud.Location;
            disp([' PCD file loaded: ' filename]);
        case '.ply'
            ptCloud = pcread(pointCloudPath);
            pointsData = ptCloud.Location;
            disp([' PLY file loaded: ' filename]);
        case '.xyz'
            pointsData = readmatrix(pointCloudPath);
            if size(pointsData, 2) < 3
                error('XYZ file must have at least 3 columns (X, Y, Z)');
            end
            pointsData = pointsData(:, 1:3);
            disp([' XYZ file loaded: ' filename]);
        case '.txt'
            pointsData = readmatrix(pointCloudPath);
            if size(pointsData, 2) < 3
                error('TXT file must have at least 3 columns (X, Y, Z)');
            end
            pointsData = pointsData(:, 1:3);
            disp([' TXT file loaded: ' filename]);
        otherwise
            error('Unsupported file format. Use .pcd, .ply, .xyz, or .txt');
    end

    end

    disp(sprintf(' Points: %d', size(pointsData, 1)));
    disp(sprintf(' X range: [%.4f, %.4f]', min(pointsData(:,1)), max(pointsData(:,1))));
    disp(sprintf(' Y range: [%.4f, %.4f]', min(pointsData(:,2)), max(pointsData(:,2))));
    disp(sprintf(' Z range: [%.4f, %.4f]', min(pointsData(:,3)), max(pointsData(:,3))));

    centerPC = mean(pointsData, 1);
    pointsNorm = pointsData - centerPC;

    scaleXY = 0.64 / (max(pointsNorm(:,1)) - min(pointsNorm(:,1)));
    scaleZ = 0.44 / (max(pointsNorm(:,3)) - min(pointsNorm(:,3)));
    scale = min([scaleXY, scaleZ, 1.0]);

    pointsNorm = pointsNorm * scale;
    pointsNorm(:, 3) = pointsNorm(:, 3) + 0.5;

    dist = sqrt(sum(pointsNorm.^2, 2));
    validIdx = dist < prctile(dist, 95);
    pointsNorm = pointsNorm(validIdx, :);

```

```

        disp(sprintf(' Point cloud normalized to %d points', size(pointsNorm, 1)));

        inputMode = 2;
        useImportedGeometry = true;
    end
end

if useImportedGeometry && inputMode == 1
    fprintf('\n--- 2D GEOMETRY IDENTIFICATION FROM IMAGE ---\n');

    edges = edge(imgGray, 'Canny', 0.15);

    se = strel('disk', 3);
    edges = imclose(edges, se);

    [B, L] = bwboundaries(edges, 'noholes');

    areas = cellfun(@(x) size(x,1), B);
    [~, largestIdx] = max(areas);
    contour = B{largestIdx};

    disp(sprintf(' Detected contour with %d boundary points', size(contour,1)));

    [rows, cols] = size(imgGray);
    contourNorm = contour ./ max([rows, cols]);

    xContour = (contourNorm(:,2) - 0.5) * 0.64;
    yContour = (contourNorm(:,1) - 0.5) * 0.52;

    t = 1:size(xContour,1);
    tq = linspace(1, size(xContour,1), size(xContour,1)*3);
    xContourUp = interp1(t, xContour, tq, 'spline');
    yContourUp = interp1(t, yContour, tq, 'spline');

    contourPoints = [xContourUp, yContourUp];
    disp(sprintf(' Upsampled contour to %d points', size(contourPoints,1)));
end

if useImportedGeometry && inputMode == 1
    fprintf('\n--- OPTIMAL SURFACE EQUATION FITTING (2D) ---\n');

    X = contourPoints(:,1);
    Y = contourPoints(:,2);

    D = [X.^2, X.*Y, Y.^2, X, Y, ones(size(X))];
    [~, ~, V] = svd(D);
    coeffs = V(:, end);
    coeffs = coeffs / coeffs(end);

    a = coeffs(1);
    b = coeffs(2);
    c = coeffs(3);
    d = coeffs(4);

```

```

e = coeffs(5);

delta = a*c - (b/2)^2;
discriminant = 4*delta;

if discriminant > 0
    centerX = (b*e - 2*c*d) / discriminant;
    centerY = (b*d - 2*a*e) / discriminant;

    trace_val = a + c;
    det_val = a*c - (b/2)^2;
    lambda1 = (trace_val + sqrt(trace_val^2 - 4*det_val)) / 2;
    lambda2 = (trace_val - sqrt(trace_val^2 - 4*det_val)) / 2;

    a_ellipse = sqrt(1 / lambda1);
    b_ellipse = sqrt(1 / lambda2);

    if abs(b) < 1e-10
        theta = 0;
    else
        theta = 0.5 * atan2(b, a - c);
    end

    disp(sprintf(' Fitted ellipse equation:'));
    disp(sprintf('  Center: (%.4f, %.4f)', centerX, centerY));
    disp(sprintf('  Semi-major axis: %.4f', a_ellipse));
    disp(sprintf('  Semi-minor axis: %.4f', b_ellipse));
    disp(sprintf('  Rotation angle: %.2f°', rad2deg(theta)));

    ellipseParams = struct('centerX', centerX, 'centerY', centerY, ...
                           'a', a_ellipse, 'b', b_ellipse, 'theta', theta);
else
    warning('Could not fit ellipse. Using circular approximation. ');
    meanX = mean(X);
    meanY = mean(Y);
    radius = std([X-meanX; Y-meanY]);
    ellipseParams = struct('centerX', meanX, 'centerY', meanY, ...
                           'a', radius, 'b', radius, 'theta', 0);
end
end

if useImportedGeometry && inputMode == 2
    fprintf('\n--- 3D SURFACE RECONSTRUCTION (POINT CLOUD) ---\n');

    disp(sprintf(' Creating alpha shape from point cloud...'));

    Xp = pointsNorm(:, 1);
    Yp = pointsNorm(:, 2);
    Zp = pointsNorm(:, 3);

    validIdx = isfinite(Xp) & isfinite(Yp) & isfinite(Zp);
    Xp = double(pointsNorm(validIdx, 1));
    Yp = double(pointsNorm(validIdx, 2));
    Zp = double(pointsNorm(validIdx, 3));

```

```

if numel(Xp) < 10
    error('Point cloud has insufficient points. Load a denser point cloud.');
```

end

```

shp_pc = alphaShape(Xp(:), Yp(:), Zp(:), 0.08);
[faces_pc, verts_pc] = boundaryFacets(shp_pc);
abdomenTri_pc = triangulation(faces_pc, verts_pc);
vertexNormals_pc = vertexNormal(abdomenTri_pc);

disp(sprintf(' Reconstructed surface: %d vertices, %d faces', ...
    size(verts_pc, 1), size(faces_pc, 1)));

Xp_full = Xp;
Yp_full = Yp;
Zp_full = Zp;

center_pc = mean([Xp, Yp, Zp], 1);
covarMatrix = cov([Xp, Yp, Zp]);
[eigVecs, eigVals] = eig(covarMatrix);
[~, idx_sort] = sort(diag(eigVals), 'descend');
eigVecs_sorted = eigVecs(:, idx_sort);

a_pc = sqrt(eigVals(idx_sort(1), idx_sort(1)));
b_pc = sqrt(eigVals(idx_sort(2), idx_sort(2)));
c_pc = sqrt(eigVals(idx_sort(3), idx_sort(3)));

disp(sprintf(' Estimated dimensions: %.4f x %.4f x %.4f', a_pc, b_pc, c_pc));
end

if useImportedGeometry && inputMode == 1
    fprintf('\n--- OPTIMAL TRAJECTORY PLANNING (2D) ---\n');
```

numPoints = 200;

```

angleArray = linspace(0, 2*pi, numPoints)';

cx = ellipseParams.centerX;
cy = ellipseParams.centerY;
a_e = ellipseParams.a;
b_e = ellipseParams.b;
theta_rot = ellipseParams.theta;

x_ellipse = cx + a_e*cos(angleArray)*cos(theta_rot) - b_e*sin(angleArray)*sin(theta_rot);
y_ellipse = cy + a_e*cos(angleArray)*sin(theta_rot) + b_e*sin(angleArray)*cos(theta_rot);

dist_from_center = sqrt((x_ellipse - cx).^2 + (y_ellipse - cy).^2);
maxDist = sqrt(a_e^2 + b_e^2);
z_ellipse = 0.5 + 0.22 * (1 - (dist_from_center/maxDist).^2);
z_ellipse = movmean(z_ellipse, 5);

t_spiral = linspace(0, 1, numPoints)';
r_spiral = 0.22 * (1 - 0.7*t_spiral);

thetaPath = angleArray + 3*pi*t_spiral;
```

```

xPath_opt = r_spiral .* cos(thetaPath);
yPath_opt = r_spiral .* sin(thetaPath);
zPath_opt = interp1(angleArray, z_ellipse, thetaPath, 'spline');

zPath_opt = max(min(zPath_opt, 0.72), 0.28);
scanPoints_optimal = [xPath_opt, yPath_opt, zPath_opt];

disp(sprintf(' Generated optimal trajectory:'));
disp(sprintf('   Waypoints: %d', size(scanPoints_optimal, 1)));
disp(sprintf('   Z range: [%.3f, %.3f]', min(zPath_opt), max(zPath_opt)));

thetaPath = thetaPath;
rPath = r_spiral;
scanPoints = scanPoints_optimal;

Xp = contourPoints(:, 1) + 0.005*randn(size(contourPoints(:, 1)));
Yp = contourPoints(:, 2) + 0.005*randn(size(contourPoints(:, 2)));
maxAxis = max([a_e, b_e]);
if isempty(maxAxis) || maxAxis < 1e-6
    maxAxis = 0.3;
end
Zp = 0.5 + 0.22 * (1 - (abs(Xp.^2 + Yp.^2)/(maxAxis^2)).^1.5);
Zp = real(Zp + 0.005*randn(size(Zp)));

elseif useImportedGeometry && inputMode == 2
    fprintf('\n--- OPTIMAL TRAJECTORY PLANNING (3D) ---\n');

    numPoints = 200;
    t_spiral = linspace(0, 1, numPoints)';

    r_max = sqrt(a_pc^2 + b_pc^2) / 2;
    r_spiral = r_max * (1 - 0.6*t_spiral);

    thetaPath = linspace(0, 3*pi, numPoints)';

    xPath_opt = center_pc(1) + r_spiral .* cos(thetaPath);
    yPath_opt = center_pc(2) + r_spiral .* sin(thetaPath);

    zPath_opt = linspace(min(Zp_full), max(Zp_full), numPoints)';
    zPath_opt = center_pc(3) + 0.1*sin(thetaPath);

    scanPoints_optimal = [xPath_opt, yPath_opt, zPath_opt];

    disp(sprintf(' Generated optimal trajectory (3D):'));
    disp(sprintf('   Waypoints: %d', size(scanPoints_optimal, 1)));

    thetaPath = thetaPath;
    rPath = r_spiral;
    scanPoints = scanPoints_optimal;

    Xp = Xp_full;
    Yp = Yp_full;
    Zp = Zp_full;
end

```

```

ur5 = loadrobot('universalUR5','DataFormat','row');
probe = rigidBody('probe');
probe.Joint = rigidBodyJoint('fix1','fixed');
probe.Joint.setFixedTransform(trvec2tform([0 0 0.12]));
addBody(ur5, probe, 'tool0');
homeConfig = homeConfiguration(ur5);

if ~useImportedGeometry
    [a,b,c] = deal(0.32,0.26,0.22);
    [u,v] = meshgrid(linspace(0,2*pi,120), linspace(0,pi/2,80));

    Xtrue = a*cos(u).*sin(v);
    Ytrue = b*sin(u).*sin(v);
    Ztrue = c*cos(v) + 0.5;

    idx = randperm(numel(Xtrue),2500);
    Xp = real(Xtrue(idx) + 0.005*randn(size(idx)));
    Yp = real(Ytrue(idx) + 0.005*randn(size(idx)));
    Zp = real(Ztrue(idx) + 0.005*randn(size(idx)));

elseif inputMode == 1
    [a,b,c] = deal(ellipseParams.a, ellipseParams.b, 0.22);
    [u,v] = meshgrid(linspace(0,2*pi,120), linspace(0,pi/2,80));

    theta_rot = ellipseParams.theta;
    Xtrue = ellipseParams.centerX + a*cos(u).*sin(v)*cos(theta_rot) - ...
        b*sin(u).*sin(v)*sin(theta_rot);
    Ytrue = ellipseParams.centerY + a*cos(u).*sin(v)*sin(theta_rot) + ...
        b*sin(u).*sin(v)*cos(theta_rot);
    Ztrue = c*cos(v) + 0.5;

elseif inputMode == 2
    a = a_pc;
    b = b_pc;
    c = c_pc;

    [u,v] = meshgrid(linspace(0,2*pi,120), linspace(0,pi/2,80));
    Xtrue = center_pc(1) + a*cos(u).*sin(v);
    Ytrue = center_pc(2) + b*sin(u).*sin(v);
    Ztrue = center_pc(3) + c*cos(v);
end

validIdx = isfinite(Xp) & isfinite(Yp) & isfinite(Zp);
Xp = Xp(validIdx);
Yp = Yp(validIdx);
Zp = Zp(validIdx);

if numel(Xp) < 10
    error('Geometry detection failed: Not enough finite points.');
```

```

end

if inputMode == 1
    shp = alphaShape(Xp(:), Yp(:), Zp(:), 0.05);

```

```

elseif inputMode == 2
    shp = shp_pc;
end

[faces,verts] = boundaryFacets(shp);
abdomenTri = triangulation(faces,verts);
vertexNormals = vertexNormal(abdomenTri);

N = size(scanPoints,1);

if isempty(N) || N == 0
    N = 200;
end

[~, uniqueIdx] = unique(verts(:, 1:2), 'rows', 'stable');
vertsClean = verts(uniqueIdx, :);

Fz = scatteredInterpolant(vertsClean(:,1), vertsClean(:,2), vertsClean(:,3), ...
    'natural', 'nearest');

scanPoints(:,3) = max(min(scanPoints(:,3), max(verts(:,3))), min(verts(:,3)));

ik = inverseKinematics('RigidBodyTree', ur5);
weights = [0.1 0.1 0.1 1 1 1];
N = size(scanPoints, 1);
configs = zeros(N, 6);
normals = zeros(N, 3);
prevConfig = homeConfig;

for i = 1:N
    [~, nearestIdx] = min(sum((verts - scanPoints(i,:)).^2, 2));
    n = vertexNormals(nearestIdx,:) / (norm(vertexNormals(nearestIdx,:)) + 1e-6);

    if n(3) < 0, n = -n; end
    normals(i,:) = n;

    zAxis = -n(:);
    xAxis_raw = cross([0; 1; 0], zAxis);
    xAxis = xAxis_raw / (norm(xAxis_raw) + 1e-6);
    yAxis = cross(zAxis, xAxis);

    pos = scanPoints(i,:)';

    T = [xAxis, yAxis, zAxis, pos;
        0, 0, 0, 1];

    [configSol, ~] = ik('probe', T, weights, prevConfig);
    configs(i,:) = configSol;
    prevConfig = configSol;
end

t = (1:N)';
tissueStiffness = 0.6 + 0.4*sin(2*pi*t/N);
probeAngle = acos(abs(normals(:,3)));

```

```

depth = scanPoints(:,3) - min(scanPoints(:,3));
attenuation = exp(-2*depth);

forces_no_opt = 2 + 1.2*sin(2*pi*t/N) + 0.4*randn(N,1);
forces_no_opt = min(max(forces_no_opt,0),5);
F_opt = 3.5;
forces_opt = F_opt + 0.6*(tissueStiffness - mean(tissueStiffness));
forces_opt = min(max(forces_opt,0),5);

conf_no_opt = attenuation .* ...
    exp(-(forces_no_opt - 3.5).^2) .* ...
    cos(probeAngle);
conf_opt = attenuation .* ...
    exp(-(forces_opt - 3.5).^2) .* ...
    cos(probeAngle);
conf_no_opt = conf_no_opt / max(conf_no_opt);
conf_opt = conf_opt / max(conf_opt);

figure('Name','Autonomous Foetal Ultrasound','Position',[50 50 1500 800]);
subplot(1,2,1);
ax3D = gca; hold on; axis equal; grid on;
trisurf(abdomenTri,'FaceColor',[1 0.75 0.7],'EdgeColor','none','FaceAlpha',0.95);
camlight headlight; lighting phong;
scanPlot = plot3(NaN,NaN,NaN,'LineWidth',2.5);
beamPlot = plot3([0 0],[0 0],[0 0],'g','LineWidth',3);

if useImportedGeometry
    if inputMode == 1
        titleStr = sprintf('Robotic Ultrasound Scan\n(From 2D Image)');
    else
        titleStr = sprintf('Robotic Ultrasound Scan\n(From 3D Point Cloud)');
    end
else
    titleStr = 'Robotic Ultrasound Scan\n(Synthetic Geometry)';
end
title(titleStr);

subplot(1,2,2);
bpmPlot = plot(NaN,NaN,'r','LineWidth',2);
ylim([90 160]); grid on;
xlabel('Scan Index'); ylabel('BPM');
title('Live Fetal BPM');

fs = 1000; tt = 0:1/fs:1;
bpmHistory = nan(N,1);

for i = 1:N
    subplot(1,2,1);
    show(ur5,configs(i,:), 'PreservePlot',false, 'Parent',ax3D);

    set(scanPlot,'XData',scanPoints(1:i,1), ...
        'YData',scanPoints(1:i,2), ...
        'ZData',scanPoints(1:i,3), ...
        'Color',[i/N,1-i/N,1]);

```



```

p = scanPoints(i,:);
n = normals(i,:);
set(beamPlot,'XData',[p(1) p(1)-0.12*n(1)], ...
    'YData',[p(2) p(2)-0.12*n(2)], ...
    'ZData',[p(3) p(3)-0.12*n(3)]);

bpm = 120 + 6*randn;
sig = sin(2*pi*(bpm/60)*tt) + 0.1*randn(size(tt));
[~,locs] = findpeaks(sig,'MinPeakDistance',fs*60/bpm*0.8);
if numel(locs)>2
    bpmHistory(i) = 60/mean(diff(tt(locs)));
end

subplot(1,2,2);
set(bpmPlot,'XData',1:i,'YData',bpmHistory(1:i));

drawnow; pause(0.04);
end

figure('Color','w');
plot(forces_no_opt,'--r','LineWidth',1.8); hold on;
plot(forces_opt,'-b','LineWidth',2.5);
yline(5,'k--','Safety Limit');
xlabel('Scan Index'); ylabel('Force (N)');
title('Force Control: With vs Without Optimisation');
legend('Without Optimisation','With Optimisation','Safety Limit');
grid on;

figure('Color','w');
plot(conf_no_opt,'--m','LineWidth',1.8); hold on;
plot(conf_opt,'-g','LineWidth',2.5);
xlabel('Scan Index'); ylabel('Ultrasound Confidence');
title('Confidence Improvement due to Optimisation');
legend('Without Optimisation','With Optimisation');
grid on;

figure('Color','w');
scatter(forces_no_opt,conf_no_opt,40,'r','filled'); hold on;
scatter(forces_opt,conf_opt,40,conf_opt,'filled');
xlabel('Force (N)'); ylabel('Confidence');
title('Force-Confidence Tradeoff');
legend('Without Optimisation','With Optimisation');
colormap turbo; colorbar; grid on;

figure('Color','w');
yyaxis left
plot(conf_opt,'g','LineWidth',2); ylabel('Confidence');
yyaxis right
plot(forces_opt,'b--','LineWidth',1.8); ylabel('Force (N)');
xlabel('Scan Index');
title('Coupled Force-Confidence Dynamics');
grid on;

```

```

fprintf('\n');
fprintf('\n');
fprintf('SIMULATION SUMMARY\n');
fprintf('\n');

if useImportedGeometry
    if inputMode == 1
        fprintf('Geometry Source: 2D Image (Contour-based)\n');
        fprintf('Image Path: %s\n', imagePath);
        fprintf('Surface Fitting: Ellipse (2D)\n');
    else
        fprintf('Geometry Source: 3D Point Cloud\n');
        fprintf('Point Cloud Path: %s\n', pointCloudPath);
        fprintf('Surface Reconstruction: Alpha Shape + Triangulation\n');
        fprintf('Cloud Dimensions: %.4f x %.4f x %.4f\n', a_pc, b_pc, c_pc);
    end
else
    fprintf('Geometry Source: Synthetic (Ellipsoid)\n');
end

fprintf('Trajectory Waypoints: %d\n', N);
fprintf('Point Cloud Samples: %d\n', numel(Xp));
fprintf('Average Confidence: %.4f\n', mean(conf_opt));
fprintf('Average Force: %.3f N\n', mean(forces_opt));
fprintf('Max Force: %.3f N (Safety: 5 N)\n', max(forces_opt));
fprintf('\n');

disp(' Simulation completed successfully.');
```

Code Explanation – Autonomous Robotic Foetal Ultrasound Simulation

This MATLAB code implements an autonomous robotic foetal ultrasound scanning simulation using a UR5 robotic arm, supporting 2D ultrasound images, 3D point clouds, or synthetic geometry as input.

A.1 Geometry Input & Configuration

The system allows switching between:

- 2D image input (ultrasound image)
- 3D point cloud input (.pcd, .ply, .xyz, .txt)
- Synthetic ellipsoidal model

User selects the geometry source via a dialog interface.

A.2 Geometry Processing

2D Image Mode:

- Image is converted to grayscale and normalized.
- Edges are detected using the Canny method.
- The largest contour is extracted and fitted to an ellipse using least-squares.

3D Point Cloud Mode:

- Point cloud is normalized, scaled, and filtered for outliers.
- Surface is reconstructed using alpha shapes and triangulation.
- Principal axes are estimated using covariance analysis.

A.3 Surface Reconstruction

A triangular mesh of the abdomen surface is created. Vertex normals are computed to determine probe orientation. This surface is used for trajectory planning and force calculations.

A.4 Optimal Trajectory Planning

A spiral scanning trajectory is generated over the detected surface. Trajectory adapts to:

- Elliptical surface (2D)
- Reconstructed point cloud (3D)

Scan depth (Z-axis) is smoothly constrained for safe contact.

A.5 Robotic Control & Inverse Kinematics

A UR5 robot model is loaded with a fixed ultrasound probe. Inverse kinematics computes joint configurations for each scan point. Probe orientation is aligned with surface normals to ensure perpendicular contact.

A.6 Physics, Force & Confidence Modelling

Simulates:

- Tissue stiffness variation
- Probe contact force
- Ultrasound attenuation with depth

Computes confidence metrics based on:

- Force deviation
- Probe angle
- Signal attenuation

A.7 Visualization & Monitoring

Real-time 3D animation of robot motion and scan path. Live simulation of foetal heart rate (BPM). Human-Computer Interface (HCI) dashboard includes:

- System status
- Force display
- Emergency stop button

A.8 Performance Analysis

Compares:

- Optimized vs non-optimized force control
- Confidence improvement

Generates plots for:

- Force safety
- Confidence trends
- Force–confidence trade-off

A.9 Summary Output

Prints a detailed simulation report including:

- Geometry source
- Number of waypoints
- Average force and confidence
- Safety validation